

19. Noise in electronics

A signal will always be contaminated by noise, either from some physical principle fundamental to the measurement process or the instrument (usually referred to as *intrinsic noise*), or from outside sources (*extrinsic noise* or, more generally, *interference*). Here, we will look at three of the most common intrinsic noise processes: *thermal*, *shot*, and *flicker*. You will encounter practical questions on noise in the last problem sheet of the course.

19.1. Thermal noise

A resistor sitting on the bench in the lab is a source of noise due to the random thermal motion of the electrons in the resistive material. If we notionally divide the resistor into two halves, then at any instant in time the left half and the right half will tend to have different numbers of electrons. Since the material has some resistance, it is able to sustain a potential difference across its ends and hence we will observe a voltage which is a function of the resistance.³¹ This type of noise was first observed by Johnson at Bell labs in the 1920s, and the effect was subsequently explained by Nyquist, hence thermal noise is sometimes referred to as *Johnson* or *Johnson-Nyquist* noise. To characterise this or any other type of noise we need to consider both its amplitude and frequency characteristics.

Amplitude characteristics. The resistor on the bench has an average noise voltage of zero. Since the voltage is caused by an ensemble of random processes, we will not be surprised to find that the amplitude distribution is Gaussian and, in this case, with a standard deviation given by

$$v_n = \sqrt{4k_B T R B}, \quad (19.1)$$

where T is the temperature in kelvin, R the resistance in ohm, k_B is Boltzmann's constant and B is the bandwidth of the measurement. The noise voltage v_n is the RMS voltage which will be measured with an instrument of bandwidth B (recall that for a Gaussian centered at zero RMS = σ). A 10 k Ω resistor measured with a really good true-RMS voltmeter with 10 kHz bandwidth will be seen to generate an RMS noise voltage of about 1.3 μ V.

Frequency characteristics. The resistor appears to generate power. The noise power generated per unit frequency is simply

$$\frac{v_n^2}{RB} = 4k_B T. \quad (19.2)$$

Equation (19.2) is the *noise power density*, i.e. the power per Hz of frequency, and since this is a constant we know that the noise has a flat spectrum, i.e. it is *white noise*.³²

³¹Note that, in an ideal conductor, while the electrons are in thermal motion, the conductor is unable to sustain an electric field and therefore there is no observable potential difference.

³²Formally, we should start from thermodynamics: the equipartition theorem is used to derive the noise power spectrum which ultimately results in equation (19.1).

This presents a conceptual problem since integrating over all frequencies implies that noise power is infinite. In practice, thermal noise is observed to be white up to tens of GHz, above which the capacitance between the two leads (tiny, but ultimately non-zero) serves to short out the very high frequencies. For practical purposes, consider the noise to be white, therefore we get a very important result: *the RMS noise voltage measured is a function of the measurement bandwidth; if we reduce the bandwidth we see less noise.*

Thermal noise model. It is helpful to think about thermal noise in terms of an ideal (noiseless) resistor R in series with a noise voltage source v_n , i.e. a Thevenin equivalent. We can also think of some stray capacitance across the whole, as above. If the resistor generates power, how can we extract this? The maximum power transfer theorem tells us that we need to connect R to another resistor R_1 of the same value. Then the voltage v_n is halved across both resistors and the power transferred to R_1 is $k_B T$ per Hz. We could imagine trying to measure this with an ideal band-pass filter of width 1 Hz and a voltmeter. However, the problem is that R_1 is also at temperature T and so it generates power $k_B T$ back into R . The *total* power in the system is still $4k_B T$ per Hz, the same as with one resistor, i.e. independent of R . Despite the impossibility of extracting the power to do useful work, the noise voltage is real and measurable, particularly when our signal is small and comes from a source with some resistance (as all real sources do). The resistive part of a source impedance generates noise, as do the resistances we use in our op-amp and filter circuits. Thermal noise usually sets the lower-limit for voltage measurements.

19.2. Signal-to-noise ratio and noise figure

This is a useful figure of merit for any measurement but we need to be sure we are comparing like with like. Generally,

$$\text{SNR}_{\text{dB}} = 10 \log_{10} \frac{P_s}{P_n} = 20 \log_{10} \frac{v_s}{v_n}.$$

P_s is the signal power ($\propto v_s^2$) which must be measured in the same bandwidth B as the noise measurement, and correspondingly the signal voltage v_s must also be an RMS measurement. For example, a 10 k Ω resistor carrying a *narrow-band* signal such as a 10 mV amplitude sine-wave will be measured by a voltmeter with 10 kHz bandwidth to have $v_s = 7.1$ mV, hence the signal-to-noise ratio is about 75 dB.

19.3. Shot noise

An electric current is not a smooth fluid flow but a flow of discrete electronic charges with some statistical variation. A DC current I will see an RMS noise current (superimposed on the DC current) given by

$$i_n = \sqrt{2eIB}, \quad (19.3)$$

where B is the bandwidth and e is the electronic charge. An analogy is rain on a tin-roof. When it is raining hard, we hear a continuous sound level (the signal). As the rain eases off, the natural statistical fluctuation in the number of drops per second (charge carriers) becomes more apparent. Hence, the variation (noise) is more apparent when the signal is small, as we can see from the signal-to-noise ratio:

$$\text{SNR} = \frac{I^2}{i_n^2} \propto I.$$

Amplitude and frequency characteristics. For a signal with a very small number of charge carriers, the statistics of the number passing a given point as a function of time follows a Poisson distribution since we assume the charge carriers are independent and random. For realistic currents, the distribution tends to the normal distribution (i.e. the amplitude distribution is Gaussian, as for thermal noise), and similarly the noise power per Hz into a load R is

$$\frac{i_n^2 R}{B} = 2eIR,$$

which is constant with frequency: shot noise is also white.

Sources. It is tempting to think that any DC current I will exhibit shot noise according to equation (19.3); in reality, in a good conductor there are long-range Coulomb interactions which act to smooth-out any variations so, to a good approximation, *there is zero shot-noise in a macroscopic current flowing in a wire or a purely passive circuit* (e.g. a battery connected to a resistor). However, where we have a semiconductor junction then charge carriers cross the junction only according to random thermal excitation and hence a diode is a source of shot noise. Shot noise was discovered in thermionic diodes by Schottky; in these vacuum devices the electrons “boil off” from a hot cathode according to Poisson statistics, then cross the vacuum to the anode. Since all modern electronics contains large numbers of semiconductor devices, shot noise is present generally, and is particularly apparent when the signal current is very small. In summary, a DC current flowing in a resistor will not generally have shot noise, however if there is already shot noise in the signal (e.g. it came from a semiconductor source) then we will see this as an RMS noise voltage $v_n = i_n R$ across the resistor.

19.4. Noise in amplifiers

The ideal amplifier is, of course, noiseless, but the real device will exhibit thermal, shot and $1/f$ noise – all of these are associated with semiconductors. In any given frequency band, the amplifier will amplify the signal presented at its input but it will also add some amount of noise. It is common to “refer this noise to the input”, as this helps us select a suitable amplifier: we compare the noise already present in the signal we want to amplify to the noise we can expect from the amplifier as it that

were also present at the input: this way we are comparing apples with apples.

It is common to use a *noise factor* when quantifying the amount of noise added by an amplifier or some other component of the circuit:

$$F = \frac{\text{SNR}_{\text{in}}}{\text{SNR}_{\text{out}}}.$$

Here the SNR is a simple ratio, not expressed in dB.³³ The ideal, noiseless amplifier has $F = 1$, since it will amplify the signal and existing noise by the same factor. A real one has $F > 1$, since the SNR at the output will be a little worse due to the noise added by the amplifier itself.

Noise can be reduced in various ways. For example, one may decrease the bandwidth of the system by selecting carefully the pass-band of the amplifier or through additional filtering (e.g. a narrow-band filter around our frequency of interest). In some applications (e.g. astronomy, but not only), it is common to operate the amplifier at very low temperatures to decrease thermal noise.

³³One may instead present a ‘noise figure’, which is F given in dB: $NF = 10 \log_{10} F$; there is also a ‘noise temperature’, which is a similar way to quantify the noise added by an amplifier – but we will tend to use the simpler *noise factor* in this course.

20. Flicker noise

Flicker noise gets its name from the “flickering”, or randomly varying current, seen in vacuum tubes. Its *power spectrum* is inversely proportional to frequency:

$$\text{power} \propto \frac{1}{f}, \quad (20.1)$$

hence, it is often referred to as $1/f$ -noise or “pink noise” (due to the fact that lower frequencies have a higher power content). Flicker noise results in equal noise power per decade of frequency, i.e. there will be the same amount of noise power in the band 1–10 Hz as there will be in the band 10–100 Hz, and so on. So:

- For high frequencies, the amount of power in the flicker noise rapidly tends to zero. Typically, the flicker noise is below other sources of noise for frequencies above a few hundred Hz;
- For low to DC frequencies, the amount of flicker noise power tends to infinity, which is problematic, but we can deal with this, as we shall see later.

Figure 20.1 shows the typical noise landscape which plagues most instrumentation systems. Flicker noise dominates up to some ‘knee’ frequency typically in the range 100–1,000 Hz. Making measurements in the kHz–MHz range is often desirable.

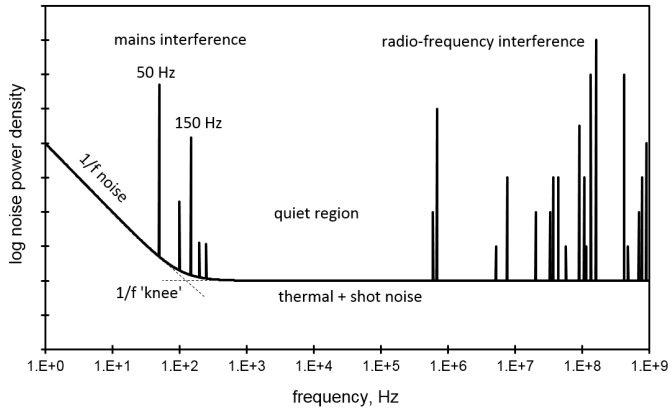


Figure 20.1: Typical noise power spectrum, showing $1/f$ noise dominating at low frequencies up to ~ 100 –1,000 Hz, mains interference at 50 Hz and its harmonics, and RF interference at high frequencies (analogue and digital radio and TV, wireless and mobile networks, etc.), leaving a relatively quiet region in the kHz–MHz region where the white noise sources (thermal and shot noise) dominate.

Thermal and shot noise are irreducible forms of noise which are generated according to well-determined physical processes. Equations (19.1) and (19.3) can be derived from well-understood physics. The thermal noise voltage on a 10 k Ω resistor is given by fundamental physical constants and quantities; it does not depend on the type of resistor. For example, the cheapest carbon-compound resistor has

the same amount of thermal noise as the most-expensive high-precision wire-wound resistor of the same value. This is not true of flicker noise, which is dependent on the type of technology in use. While flicker noise seems to be a universal phenomenon associated with all measurements, we will use the example of the resistor to illustrate. Imagine a 1 k Ω resistor carrying a constant of 1 mA applied from a really good, steady source. Assume for the moment that there is no thermal or shot noise. Then, if we measure the the RMS flicker noise voltage across the resistor in, say, any decade of frequency, we will find different results for different resistor types – see Table 20.1. This is extraor-

Table 20.1: Flicker noise according to resistor type.

Resistor technology	Flicker noise voltage
Carbon composition	1 μV_{RMS}
Carbon film	0.2 μV_{RMS}
Metal film	0.1 μV_{RMS}
Wire-wound	<0.1 μV_{RMS}

dinary, given that we assume the current supplied to the resistor is constant, so the voltage should be too (assuming no shot/thermal noise). There is no satisfactory universal physical explanation for this effect, which gives rise to another name: *excess noise*; it is likely that there are multiple phenomena which cause this type of noise in any system (e.g. the nature of the electrical contacts), rather than a universal explanation.

Flicker noise can be understood by imagining that the *resistance* of the resistor fluctuates, hence generating the noise voltage. This “understanding” (as it is not physically-based) is useful as it allows us to reason why different resistor technologies show different levels of noise. It seems that details of the construction of the resistor influence the overall amount of noise, such as the type of resistive materials used and how the leads are connected to it.

The flicker noise power in a bandwidth $B = f_1 \dots f_2$ is

$$P = k \int_{f_1}^{f_2} \frac{df}{f}, \quad (20.2)$$

where k is some constant which depends on the technology.

Flicker noise is seen in almost all electronics applications, and is particularly apparent in any kind of semiconductor device, where the idea that a junction has some resistance is also useful for understanding the effect. However, flicker noise seems to be a more fundamental property of any measurement, and indeed a fundamental property of nature itself. If we record the speed of ocean currents (for example) over a long period of time and then take a frequency spectrum from our time-series data, then the result tends to have a $1/f$ power spectrum. What this tells us is that the longer-period variations carry relatively more power in the signal than the faster variations. This pattern is repeated across nature. Another example is that the loudness of a reasonably long and complex piece of

music (say a symphony) tends to have $1/f$ characteristics. Since the power increases as frequency decreases, the question becomes: how low can we go in measurement frequency? According to equation (20.2), the total integrated power between DC (zero Hz) and 1 Hz tends to infinity, which seems to imply that any truly DC measurement is impossible. This is generally true, since any true DC measurement of any quantity would take an infinite amount of time to complete. In this sense, there is no such thing as a truly constant parameter (such as voltage, current, temperature,...), we can just constrain the lower frequency of our measurement. Consider another oft-cited example of $1/f$ behaviour: the level of water in the Nile. This is probably the single quantity with the longest history of recording: there are measurements of the water height every day going back about 4,000 years. The lowest frequency in the data is therefore about 4×10^{-12} Hz. This is a very low frequency indeed, but not constant or DC. Thus, measurement-time is an effective lower-limit on frequency, which serves to keep the flicker noise finite.

On a log-log plot of flicker noise *power* vs. frequency such as that in Figure 20.1, the flicker noise falls-off at -10 dB/decade, while we have seen that thermal and shot noise are constant with frequency. Therefore, a typical overall noise plot for any measurement will show $1/f$ dominating at low frequencies, and thermal/shot at higher frequencies. An important consequence of flicker noise is that low-frequency measurements are harder to do than higher frequency measurements.

21. The lock-in amplifier

The *lock-in amplifier* is an application of the *phase detector* which we looked at in Section 13. Lock-in techniques can be used to achieve extraordinary levels of noise rejection through simultaneous application of two techniques:

1. *Bandwidth narrowing*, which as we have seen is the best way to minimise noise in our measurement;
2. *Modulation* of a low frequency, or essentially ‘DC’, up to a higher frequency where it will be less affected by flicker noise.

A very important implication of equation (13.3) is that the difference between the two input frequencies has to be very small if we are to get any output from the phase detector. We can set the value of ω_c by choosing the values of R and C in the output filter, and therefore we can make the range of frequencies that the circuit responds to very narrow indeed (values as low as 1 Hz are possible). Since the reference frequency may be kHz or MHz, this means the circuit is very good at “picking-out” frequency components of the input signal with very high *spectral (frequency) resolution*. Generally, our input signal will consist of many frequencies (and noise) so the phase detector is good for asking questions such as “What is the amplitude of the input signal at 1 kHz measured in a bandwidth of 1 Hz?”.

A typical commercially-available implementation of the phase detector is known as the *lock-in amplifier* and in its simplest implementation this is a box with an input, two control knobs and an output. The first control is used to set the reference frequency (an internally generated square wave) and the second allows the phase of the reference to be adjusted. The procedure to be used goes as follows: the input is connected to the signal we wish to study and the output V_{out} is usually sent to an oscilloscope. The reference frequency is set to 1 kHz and if there is any component of the input signal at this frequency then we will see an output on the scope. According to equation (13.3) we may find that the output varies a little so we fine-adjust the frequency knob to get a fix (or ‘lock’) with a steady output. At this point $\Delta\omega = 0$, so now equation (13.1) comes into play. We now adjust the phase knob to get a maximum output, and from this we can determine the value of the amplitude A at the given frequency.

Practical lock-in amplifiers also have complicated extra features but the only one we need to be concerned with is the ability to take the reference frequency as an external input so we can generate this signal ourselves.

21.1. Signal analyser

The lock-in amplifier gives us the capability to measure the amplitude of a signal at a specific frequency and with a very narrow bandwidth. This is the basic requirement of a *signal analyser*, a device which gives as output the amplitude of a signal as a function of frequency. We can use a signal generator to sweep the reference frequency over the required range and we can measure the amplitude

at the output as a function of frequency. This will give us the amplitude spectrum of the signal. Note that, in practice, signal analysers – such as the one you have used with the ELVIS equipment – use a different technique; however, it is important to understand the concept, and in fact some high-end *spectrum analysers* do use a variation on the method described above.

21.2. Extracting signal from noise

One of the most powerful uses of this technique is when we have a signal we wish to measure at a known frequency, but heavily contaminated with noise. For example, we may have a sensor which we know produces a signal at 1 kHz but with a very small amplitude, say tens of μV . We may find that our signal has noise of some few volts RMS (measured in a bandwidth of say 10 kHz). The signal to noise ratio is then some -100 dB. The situation seems hopeless: were we to plot the data in the time-domain we would just see the noise; the signal at 1 kHz would be pretty well buried. However, recalling that noise is usually broad-band, we understand that we can reduce the amount of noise in a measurement by reducing the bandwidth of the measurement. For example, for thermally-generated noise we found that $v_{n,RMS} \propto \sqrt{B}$. The lock-in amplifier can recover the signal because:

1. We expect a well-defined signal frequency which is repetitive and can be synchronously detected (in comparison with the noise which is random and ultimately averages to zero);
2. The lock-in technique is highly selective in frequency, therefore it reduces the bandwidth of the measurement.

In practice, a SNR of -100 dB or worse is manageable with this technique.

Bandwidth limiting

Consider a measurement where the observed noise is 10^6 times larger than the signal. If the noise is principally thermal in origin, then the noise per unit Hz is a constant so we can reduce the total noise measured by reducing the bandwidth of the measurement. Measuring with an oscilloscope (typical bandwidth 100 MHz or more) we would be unable to see the signal. However, if we first filter the signal with a band-pass filter of width 100 Hz, then the SNR improves by a factor of a million. Note that the part of the signal we are interested in must of course be within the filter range.

In summary, we should think of this as a frequency-domain technique. If the signal of interest has a definite and fixed frequency (i.e. it is *narrow-band*), and if the noise is *broad-band* (i.e. across the whole frequency range of the measurement), then we can see that in the frequency-domain the two sources look quite different, and can be separated by the lock-in technique.

21.3. Low-frequency or DC signals

This approach works well for AC signals with a frequency that it is sufficiently high that it is above the regime where the $1/f$ noise dominates. Under these circumstances, the noise – measured in the same band as the signal – is usually tolerable. However, as the signal frequency reduces, and ultimately as we go towards a ‘DC’ measurement, then the $1/f$ noise dominates and can overwhelm the small signal.

We may have a detector which gives a constant voltage of a few μV with some volts RMS of noise. Another example would be measuring the intensity of light from a distant source, when the detector output is swamped by ambient light. Or we could be trying to detect faint radio-emission from a star. These measurements are all small, constant values subject to noise. The solution here is to shift the frequency of the signal to be measured from near DC to a higher frequency (well above the regime where the $1/f$ dominates).

This can be done by a variety of methods which we will look at next, though they are all essentially *modulation techniques* or, as it is more typically known for this application, “chopping”.

21.4. Modulation techniques

We can move a DC or very low-frequency signal to a higher frequency by modulating or chopping the signal. The name derives from optical experiments where a ‘chopper’ wheel (a rotating blade) physically interrupts the light beam in front of the detector. This modulates the light beam from a constant to an on/off signal at a well defined frequency (given by the rotation rate of the chopper wheel). In Fourier terms, the signal changes from a delta function at zero frequency to a delta function at the modulation frequency. This has the added advantage that we can use the same frequency source for the modulation and the reference of the lock-in amplifier, so we are guaranteed to have the precise same frequency. This is best illustrated with an example.

Demonstration experiment using light intensity. Imagine we have an LED powered by a battery, and on the other side of the room we have a photodetector. The question is: can the detector tell whether the light is on or not? In a very dark room, with not too large a distance between the two, it is possible. With the LED off, the trace on the oscilloscope is a noisy straight line, and when we switch the LED on then the line goes up just a little. Note here that the detector is acting as an intensity meter. The LED light is a constant output ‘DC’ signal. The detector responds to light of all visible wavelengths, so when we open the curtains the detector is flooded with ambient light millions of times brighter than the LED. We have to drop down the scale of the scope by several orders of magnitude to get a new reading of the intensity. Now, if we switch the LED off/on we see no difference in the trace because it is simply buried by the ambient light.

Now imagine we discard the battery and instead power the LED with a square wave from a signal generator. This modulates the light intensity from DC to the frequency of the signal generator. Further, we also connect the signal generator output to the reference input of a lock-in amplifier, and the output of the detector to the signal input of the lock-in. Now we can synchronously detect the LED intensity at the modulation frequency. Because modulation and reference frequencies are the same, we just need to adjust the phase knob to get a maximum output. We will see a nice strong constant signal out of the lock-in. If we disconnect the LED it goes to zero. This works the same both in the dark and in bright ambient light.

This is a striking example of extracting a tiny signal from noise. To be clear about what is happening here, we need to recall that

1. The modulated signal has a very narrow, well-defined (fundamental) frequency;
2. The noise has lots of sources: mostly the more-or-less constant ambient light, plus some 50 Hz from mains lights, plus $1/f$ and thermal noise contributions from the detector system;
3. Modulation moves the signal above the low-frequency noise, to where we have only broad-band thermal noise, etc.;
4. Consequently, the noise in the 1 Hz bandwidth of the lock-in is actually quite small.

Chopping. The demonstration experiment above is not very realistic, since it is usually impractical to electrically modulate the signal without introducing significant systematic error. Usually, the quantity to be measured needs to be mechanically or optically ‘chopped’. The name derives from optical experiments where a rotating wheel or blade physically interrupts a light beam and turns it from a constant to a modulated quantity. Since we know the frequency of the ‘chopper’ wheel we can use this as the reference for the lock-in. Typically the arrangement is similar to that described above except here the signal generator drives the chopper.³⁴

21.5. An example from radioastronomy

A pleasing example of the chopping technique comes from the world of radioastronomy. A radio-telescope is a dish antenna used to measure radio-frequency emission from distant stars. The signal is very faint, while noise sources abound: we have the general cosmological background, terrestrial (man-made) radio-sources, plus all the usual electrical noise sources in the receiver. To make this worse, fluctuations in the atmosphere and ionosphere add extra low-frequency noise to the signal. Radioastronomers use the chopping/synchronous-detection scheme as

³⁴You can watch a good explanation of the chopper-plus-lock-in amplifier here: <https://www.youtube.com/watch?v=SwPt9grlsME>.

described previously. This can be done by physically rocking the dish side-to-side thus sweeping it across the target star. This directly modulates the signal intensity, while all the noise sources remain the same. With large dishes this is impractical, so often there is a secondary reflector at the focus of the dish (Cassegrain type), and it suffices to rock the secondary. A further refinement, which avoids any mechanical action (and can be run much faster) is a setup which uses a single dish with two signal-receiving waveguide/antennae. One is exactly on-axis and receives the target signal, the other is slightly off-axis so it receives all the noise sources but none of the signal from the star. The radiometer (powermeter for radio-frequency signals³⁵ is electrically switched between the two sources at the modulation frequency.

³⁵Ultra-sensitive radiometers use another noise minimisation technique we have already discussed: the receiver/amplifier stages are cooled to cryogenic temperatures in order to minimise thermal noise.